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The overlooked contribution of trees outside forests to tree cover and woody biomass across Europe

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ABSTRACT

Trees are an integral part of almost all European landscapes, but only forest resources are systematically assessed by national inventories, and the extent to which trees in urban and agricultural areas contribute to biomass and carbon stocks at the national level remains largely unknown. Here we make use of nanosatellite imagery to generate canopy cover, height, and above-ground biomass maps for the entire European continent from 3-m resolution imagery. Our country-scale biomass estimates have a systematic bias of only 7.6% (overestimation; $R=0.98$) when compared to national inventories of 30 countries, and our new dataset is sufficiently highly resolved spatially to support the inclusion of tree biomass outside forests, which we quantify to 0.8 Pg. Although this represents only 2% of the total European tree biomass on average, large variations between countries are found (for example 10% for UK) and trees in urban areas contribute substantially to national carbon stocks (for example 8% for the Netherlands). The good agreement with national inventory data, the scalability, and the high level of detail across landscapes, including trees outside forests, make our approach attractive for operational implementation to support national carbon stock inventory schemes.

33 Introduction

34

35 The quantification of wood resources at national as well as at local scale is a prerequisite for sustainable management
36 of timber resources, habitats, and carbon stocks [1-5]. Local forest management practice and planning relies to an
37 increasing extent on national maps of forests and forest resources, and the accuracy of national carbon stock estimates
38 may be largely improved by inclusion of such maps at regional and national levels [6,7].

39

40 Most countries focus their inventories and mapping on closed canopy forests [8-10]. Trees outside forests are often
41 associated with scattered dryland trees in arid regions where rainfall is not sufficient for trees to form closed canopies
42 [11]. However, in northern countries a substantial part of the woody resource may be growing in hedgerows, in
43 gardens, parks, urban areas, grasslands and in agricultural lands. These trees outside forests contribute to carbon
44 storage, provide resources for local communities, they modify the local climate, are an important part of habitat
45 networks, impact the hydrological cycle, and thus represent an important economic and social value [12-14]. Many
46 European countries comprise large agricultural and urban landscapes and the exclusion of trees outside forests from
47 systematic carbon stock assessment potentially implies a bias in national inventories, and also impacts scientific
48 studies and climate models using only closed forest areas as input [15,16]. For example, reports from the UK have
49 shown that woody resources provided by trees outside closed canopy forests can exceed those provided by forests
50 [17].

51

52 Consistent monitoring of both forest and non-forest trees at national and continental scale remains a challenging
53 endeavor. This is because contemporary forest maps derived from medium resolution (10-30 m) satellite imagery
54 usually misses non-forest trees and do not allow for a clear definition and separation of forest and non-forest areas
55 [18]. Other maps merge all types of trees under the variable “tree cover” or “forest cover” [19,20] and it often remains
56 unclear which size classes of trees and shrubs are included and which are excluded. Certain guidelines provide
57 consistent forest definitions, for example by the United Nations Food and Agriculture Organization (FAO) [21].
58 However, the quality and type of data available at national and continental scale is often not sufficient to apply the
59 guidelines consistently in remote sensing studies. Only few countries provide high quality aerial imagery at national
60 scale that allow for an assessment of all forests and trees [22,23], but the inclusion of canopy height data derived from
61 airborne lidar campaigns is decisive, to separate large trees from shrubs and bushes, and to estimate the wood volume,
62 dry mass, and carbon stocks. This is particularly the case for managed forest areas, where small and privately-owned
63 lots characterized by different management strategies, cause a high spatial heterogeneity within areas considered as
64 closed canopy forest [5]. High spatial resolution tree cover maps are not sufficient for a quantification of wood
65 resources as the tree height is unknown. Existing global and continental scale canopy height [24-26] and medium
66 resolution biomass maps [27,28] from space-borne sensors largely ignore trees outside forests, providing an
67 incomplete assessment of resources. National canopy height data at a high spatial resolution (<5 m) is expensive and
68 often only available for parts of the country, if at all. It thus remains unknown to which extent trees outside forests
69 across the European continent represents a hidden and undervalued resource, as it is the case in the UK [17].

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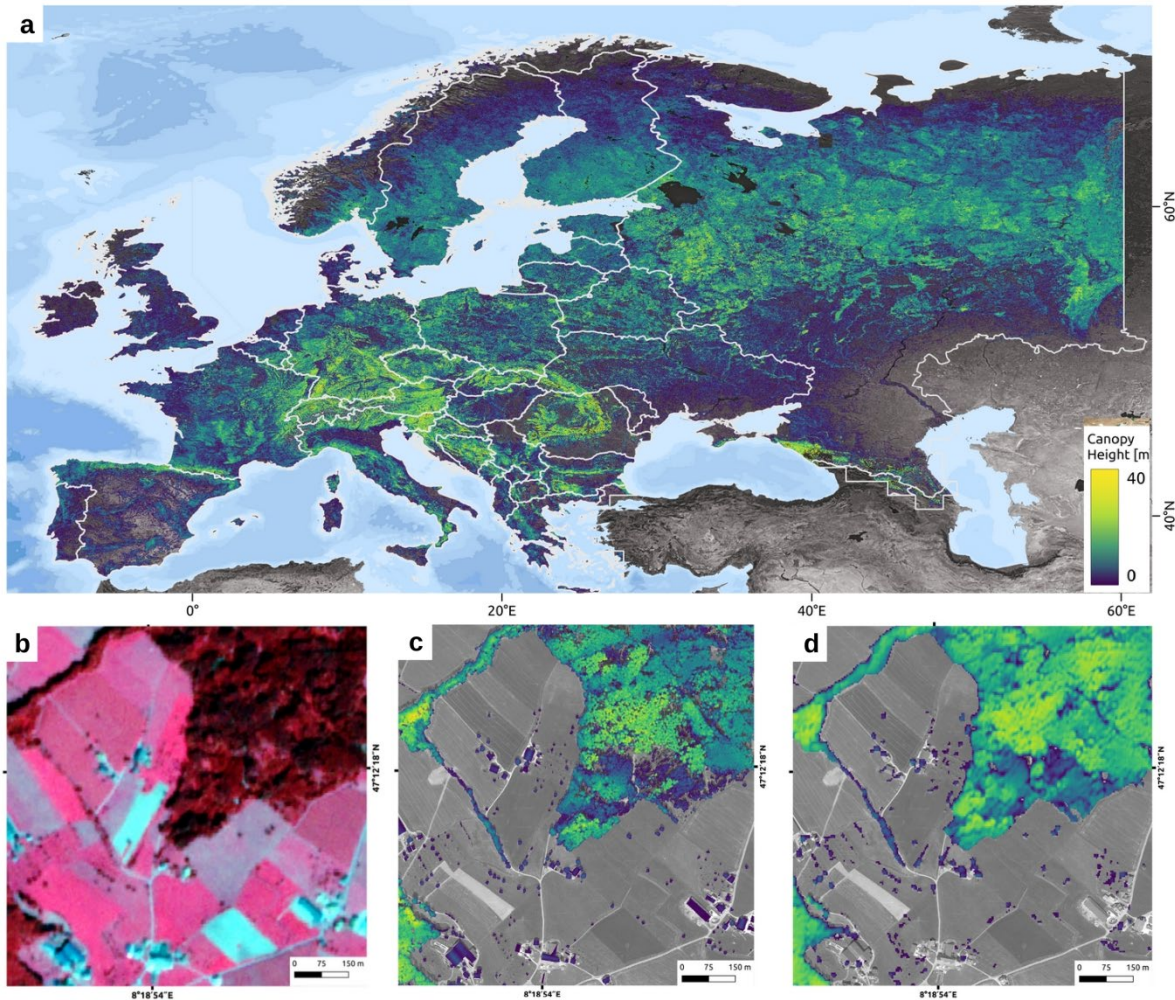
71 Here we present continental-scale maps of forests, woodlands, and trees outside forests, including their height at 3 m
72 resolution in 2019, at a level of details otherwise only previously accessible from the use of airborne lidar surveys. We
73 retrieve these data from cost-efficient nanosatellites available at a daily basis and at a spatial resolution high enough to
74 map large individual trees (>3 m height) using a deep learning approach where a convolutional neural network (CNN)
75 is learned end-to-end from airborne LiDAR reference data. Furthermore, we use inventory data to convert tree cover
76 and height into and above-ground biomass maps including both forest and non-forest tree biomass, forming the basis
77 for quantifying and comparing forest and non-forest tree resources across European countries. We further apply the
78 model trained in Europe to the boreal and temperate zones of North America to test the scalability of the approach to
79 different areas evaluated on independent datasets.

Results

A canopy height map at 3-m resolution for Europe

We used 0.7 million km² of canopy height data from airborne LiDAR available for Denmark, Netherlands, Switzerland, Estonia, Spain, Finland, and Wales (Extended Data Fig. 1b) to train a deep learning model and predict tree height and binary tree cover from PlanetScope imagery in 2019 at 3 m resolution for all Europe (Fig. 1; 10 million km²). The training data cover boreal, temperate, and Mediterranean areas (Extended Data Fig. 1a). Model performance was evaluated on 10,000 km² of randomly sampled airborne LiDAR data (organized as 1x1 km tiles), which was not used for model training and hyper-parameter optimization (Fig. 2; Extended Data Fig. 2).

At the pixel-level, canopy height was predicted with a low random error for boreal (RMSE = 4.33 m, relative RMSE (rRMSE) = 21.7%) and temperate zones (RMSE = 5.06 m, rRMSE = 27.8%), but the diverse tree structure of Mediterranean regions results in a higher uncertainty (RMSE = 6.44 m, rRMSE = 32.2%), observed as a more severe underestimation at higher canopies (> 30 m height) (Fig. 2a), which can be partially explained by the uneven distribution of the validation data, with a lower number of points for higher canopies (Fig. 2b). Aggregating tree cover and height to 1x1 km grids, we found that the overall systematic bias (see Methods for equation) between our PlanetScope predictions and LiDAR data for areas outside forests, here defined as trees that we mapped in the ‘urban’ and ‘cropland’ classes using a global land cover land use dataset (GLCLU) [29], was -1.6% for canopy height and -2.6% for canopy cover. For closed canopy forests, here defined as areas mapped as forest in the GLCLU dataset, the bias was -1.2% for canopy height and +0.4% for canopy cover (Fig. 2c). When comparing two global canopy height products with airborne LiDAR derived canopy cover and height, the systematic bias was high, in particular for trees outside forests: Here the bias for canopy height was +15.4% and for canopy cover +231.2% for a fused Sentinel-2 and spaceborne LiDAR (GEDI) based map at 10-m resolution [26], while it was -31.3% and -95.8% for canopy height and cover, respectively, for a Landsat- and GEDI-based map at 30-m [24] (Fig. 2d).



104

105 **Figure 1. PlanetScope based canopy height map for Europe at 3-m resolution.** **a**, Spatial distribution of average canopy
106 height aggregated to 1-km resolution for visualization purpose. **b**, PlanetScope imagery example in Switzerland displayed
107 as near infrared, red and green false color composite. **c**, Corresponding canopy height models (CHM) from airborne

LiDAR at 0.5 m resolution. **d**, Same area as **b**, but showing the PlanetScope-based canopy height prediction at 3 m. The land basemaps in **a**, **c** and **d** are from Google Maps satellite imagery (Imagery © 2022 CNES / Airbus, Landsat / Copernicus, Maxar Technologies, Map data ©2022). The ocean basemap in **a** is from www.naturalearth.com.

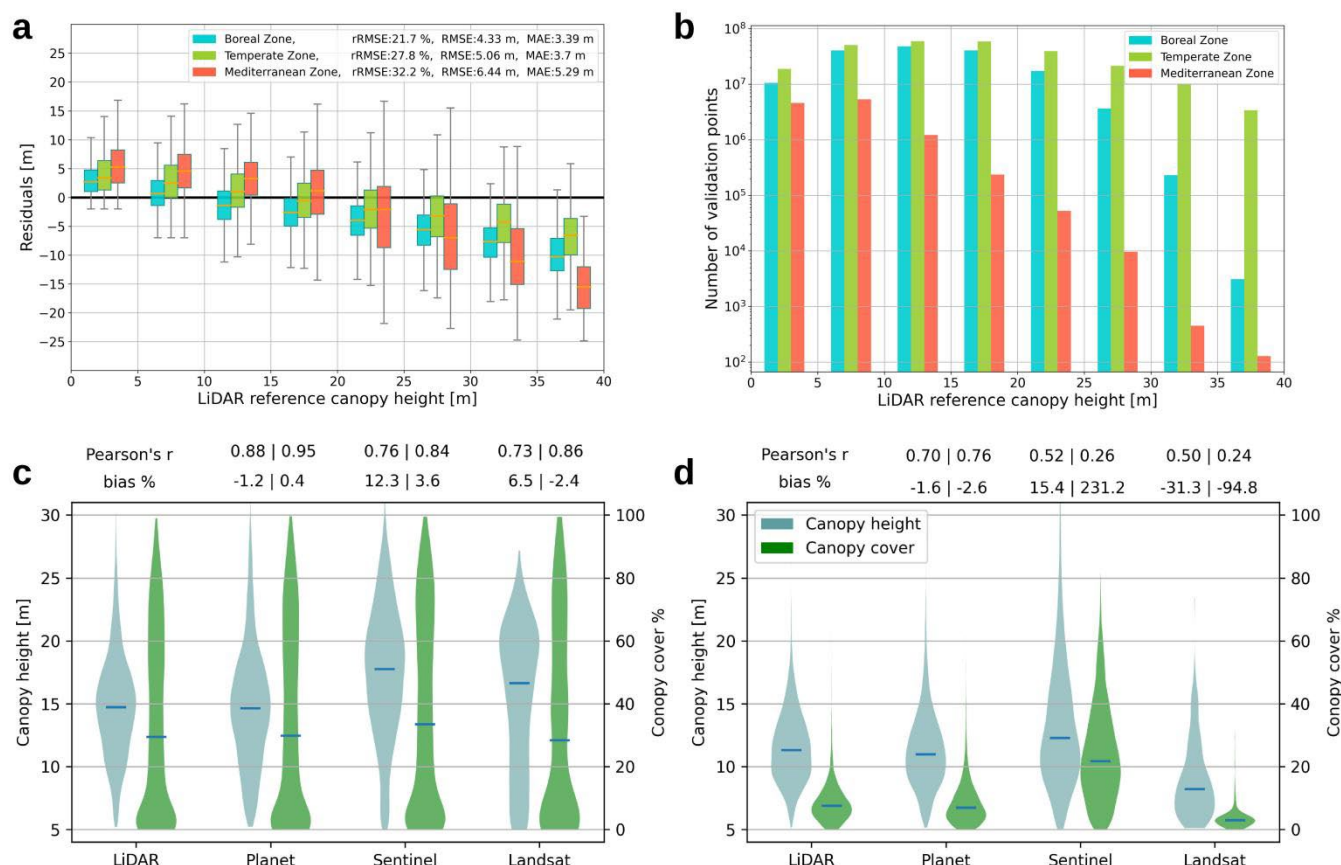


Figure 2. Evaluation of canopy height and cover. **a**, Residuals at 3x3 m between aerial CHM and our PlanetScope-based predictions, grouped in 5 m height intervals. **b**, Number of evaluation points used in **a**. See Extended Data Fig. 1 for biome zones [30]. **c-d**, Violin plots of canopy height and canopy cover for airborne LiDAR based CHMs, PlanetScope predictions (this study), a Sentinel-2 based product from ref [26], a Landsat based product from ref [24] aggregated to 1x1 km samples (that were not used for training and parameter optimization) divided into forest areas ($n = 4,772$) (**c**) and non-forest areas ($n = 5,228$) (**d**). Forests and non-forest areas are defined by the GLCLU dataset. Two-tailed Pearson correlation and bias are calculated between the airborne LiDAR based CHM samples and the other products (all aggregated to 1x1 km). See corresponding scatter plots in the Extended Data Figs 2,3. The horizontal blue line is the median and the violins show the data distribution. See Methods for the bias calculation.

Quantification of tree cover outside forests across Europe

To refine the forest vs. non-forest area definition in relation to our dataset, we used the FAO definition [21] to separate forests from non-forest trees. Here, we aggregated our PlanetScope-based tree cover and tree height maps into 0.5-hectare grid cells. Following the FAO definition, woody vegetation in 0.5-ha grids where the canopy cover of trees taller than 5 m height exceeded 10%, and the land use was not agricultural land or urban according to ref [29], was considered as forest. Note that the FAO definition includes areas of trees that potentially can grow above 5 m, which cannot be resolved for our map. Conversely, trees in 0.5 ha grids where the cover of trees taller than 5 m was below 10%, or the land use was urban or agricultural land, were considered as trees outside forests. Following these definitions, we find that forests cover 377 million hectares, equivalent to 28% of the total land area shown in Figure 1. The canopy cover of trees outside forests represents 16.1 million hectares, which is 1% of the total land area and 4% of the total tree cover. The precise percentage of cover due to trees outside forests depends on a number of factors including land cover maps, and spatial resolution of the product (Extended Data Fig. 4).

To analyze the spatial distribution of both forest and trees outside forests, we aggregated our results at country scale (Fig. 3a,b) and to one degree units (Fig. 3c). Results show that almost half of all countries (44%) have a national forest cover between 30-45%, and trees outside forests contribute less than 5% to the total tree cover for about half (53%) of the countries. Northern Europe is mostly covered by forest with limited trees cover outside forests: Finland, Slovenia, and Sweden have the highest percentage of national forest cover (66%, 63%, and 61%) and the lowest contribution of tree cover outside forests to the total tree cover (1.8%, 2.8%, 1.9%, respectively). On the contrary, the least forested countries (United Kingdom: 12%, Netherlands: 12%, Denmark: 15%) have the highest contribution of tree cover outside forests to

the total tree cover (22.1%, 24.5%, 19.5% respectively). The highest contribution of trees outside forests at one degree grid-scale is found in western Europe along the coast and southwestern Europe (up to 25% in Fig. 3c,e) where agricultural and residential areas are concentrated and forest coverage is consequently low. The histogram results show a roughly even distribution of forest cover percentage, with most of the trees outside forests accounting for less than 10% of the total tree cover (Fig. 3d,e). For forest areas, our results compare well with national FAO statistics (Extended Data Fig. 5). Differences can be explained by the slightly different definitions of forests. For example, the FAO definition includes areas with shorter trees that can grow taller than 5 m, such as reforestation and afforestation, which was excluded in our map.

We then quantified tree cover for different land use/cover classes (Fig. 3f), by aggregating our data into 0.5 ha grids. For Europe as a whole, we identify 15.5 million ha of trees located outside the “forest” class of GLCLU, half of them is found in “dense short vegetation” (hereafter referred to as grassland), with a median (25-75 percentiles) canopy cover of 7.2% (2.1-19.2%). Urban areas have on average the highest canopy cover (12.7%, 4.5-26.9%), with 5.05 million ha aggregated tree cover. A total of 2.67 million ha tree cover is found in cropland with a rather low canopy cover percentage per ha (4.6%, 1.4-12.2%). Detailed statistics are found in Extended Data Table 1.

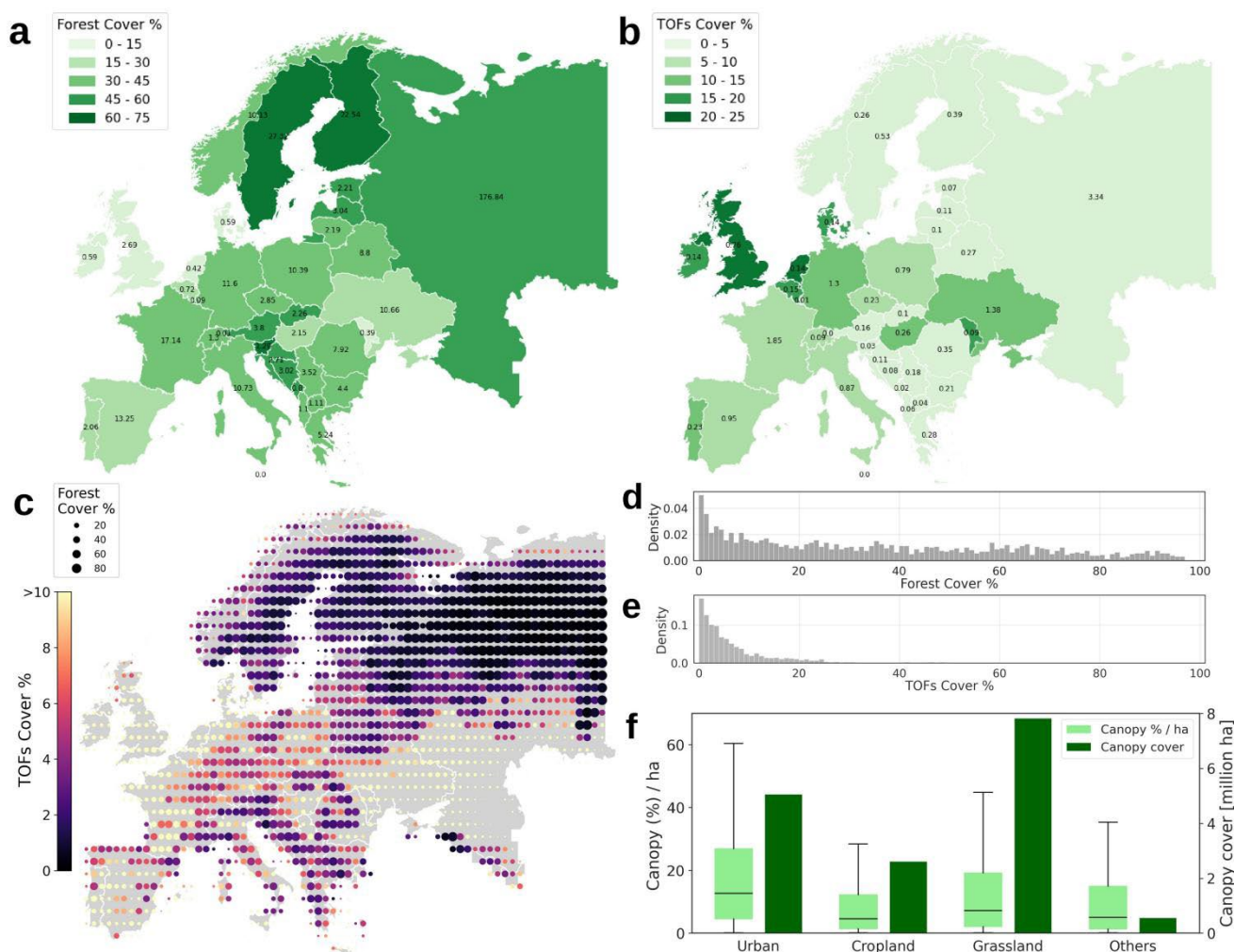


Figure 3. Quantification of tree cover outside forests across Europe. **a**, Total forest area in million hectares and the percentage of surface area at country scale. **b**, TOFs area in million hectares and the percentage of total tree cover at country scale. **c**, Percentage of forest (size of the circle) and TOFs (color of the circle) cover for 1x1 degree grids. **d**, Forest cover percentage histogram at 1x1 degree as shown in **c**. **e**, TOFs cover percentage histogram at 1x1 degree as shown in **c**. **f**, Canopy cover of different GLCLU landscape types: the boxplots show the canopy cover percentage per ha and the bars denote the total canopy cover area per class. For the boxplots, the start of the horizontal line represents the minimum value, vertical lines represent first quartile, median and third quartile values respectively, and the end of the horizontal line represents the maximum value.

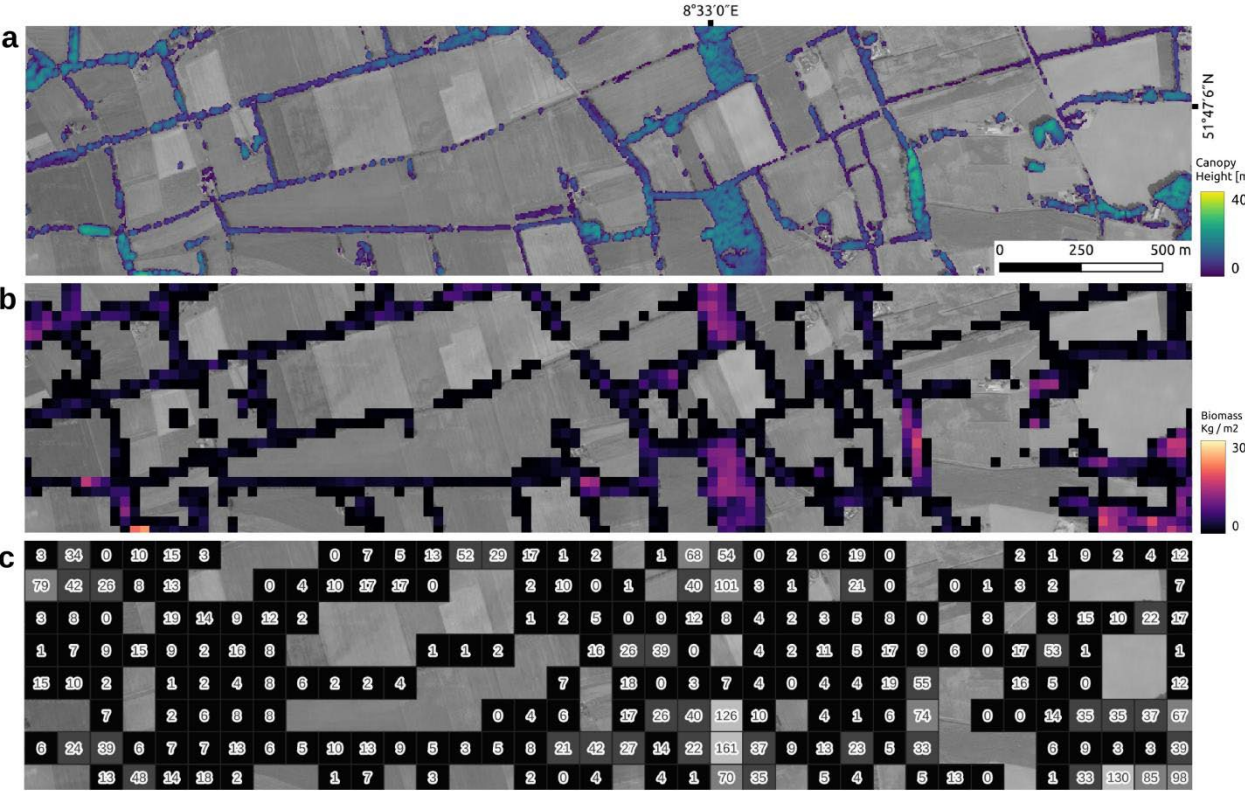
Quantifying above-ground biomass across Europe

To quantify the woody resources into above-ground biomass, we used plot based National Forest Inventory (NFI) data and airborne LiDAR CHM data from Denmark [33] to establish allometric relationships between above-ground biomass

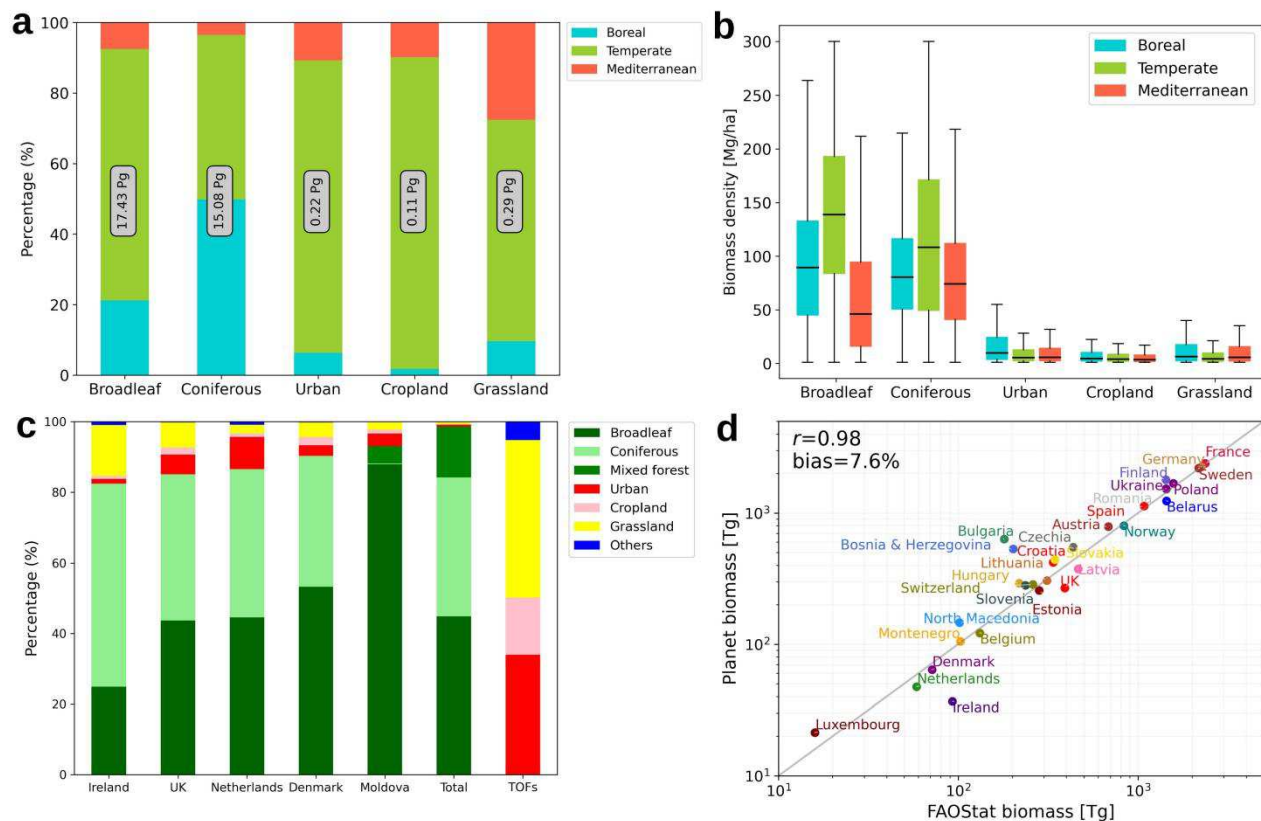
and canopy height, both averaged at plot level (n = 11,296). Separate relationships were derived for broadleaf forest (n = 7,232; bias = -8.8%), coniferous forest (n = 3,768; bias = -6.5%), mixed forest (n = 7,536; bias = -9.9%), and plots with sparse tree cover (n = 409; bias = -5.9%) (Extended Data Fig. 6) using the Copernicus map [32] for the separation (see Methods). We aggregated above-ground biomass to the hectare level (100x100 m) including both forest and non-forest trees (Fig. 4). The overall uncertainty is the combined uncertainty due to the canopy area and height prediction, and to the height to biomass conversion, and was quantified by comparing our final above-ground biomass product (after application of the allometric conversion on PlanetScope canopy height and cover data) with field measured biomass from the Danish NFI data at plot-scale (n = 3,451) and country scales (n = 30). At plot scale, the correlation was moderate (r = 0.53, bias = -23%) (Extended Data Fig. 7c), which was expected due to a large uncertainty related to both satellite and field data, such as geolocation and image quality errors. Importantly, these errors were found to be not systematic, as demonstrated by a comparison of statistics aggregated to country scale, both from NFI and satellites. Here, the systematic bias was -10% (underestimation) for Denmark, which can be explained by the fact that clearcut areas and young tree plantations are part of the national statistic, but are not mapped in our product. The bias was +7.6% over 30 countries with a Pearson correlation of 0.98 (Fig. 5d; Extended Data Fig. 8a). The relatively even distribution of the error across countries demonstrates the transferability of the approach beyond Denmark. A previously published state-of-the-art biomass map from reference [28] had a larger bias of +17.3% when compared against country statistics. Aggregated to 1x1 km scale, our map compares reasonably well with existing products (Extended Data Fig. 7d,e,f) [28,31,27], with the advantage of including information on non-forest trees (Fig. 4).

Following the FAO forest definition, European forests have an overall above-ground biomass of 35.2 Pg, and trees outside forests represent only 0.8 Pg, which is a proportion of 40:1. By further aggregating our biomass maps into forest types and biomes, we found that broadleaf and coniferous forests in the temperate zone have a biomass density of 140.4 (84.8-195.8) Mg/ha and 110.1 (50.4-174.6) Mg/ha, respectively (Fig. 5a,b). Using the GLULC land cover classification, our results reveal that trees in urban areas have a total biomass of 0.22 Pg with a median biomass density of 5.6 (2.5-13.2) Mg/ha, trees in croplands have 0.11 Pg with a median biomass density of 3.8 (1.9-8.5) Mg/ha, and trees in grassland have 0.29 Pg with a median biomass density of 4.6 (2.1-11.1) Mg/ha. Overall, 44.6% of the tree biomass outside forests is found in grasslands, 34.0% in urban areas, and 16.2% in croplands (Fig. 5c).

At country scale, we ranked the top 5 countries having the highest contribution of non-forest trees for different land use/cover classes (GLULC) to the national biomass stocks. Ireland was ranked first, with 16.5% of the tree biomass located outside forests, followed by the UK (14.8%), the Netherlands, which has 8.2% of its national biomass located in urban areas, and Denmark (9.7%) (Fig. 5c). We found a negative relation between the total tree cover percentage and the biomass contribution by trees outside forest (Table. 1). The Netherlands stands out with both the highest tree cover outside forest (24.6%) as well as a high biomass contribution in percentage (12.2%). France has the largest total biomass (2,388.2 Tg), while the tree biomass outside forests is only of 77.2 Tg.



208 **Figure 4. Above-ground biomass estimation using PlanetScope.** **a**, PlanetScope-based canopy height map at 3 m
 209 resolution. **b**, Biomass density estimation based on **a** derived from allometry equation at 30x30 m in kg per m². **c**, Biomass
 210 density per hectare aggregated from **b** in the unit Mg/ha. The basemap is from Google Maps satellite imagery (Imagery ©
 211 2022 CNES / Airbus, Landsat / Copernicus, Maxar Technologies, Map data ©2022).



212
 213 **Figure 5. Above-ground tree biomass across tree types, land use/cover and countries.** **a**, Total tree biomass for
 214 different forest and land use/cover types, the color reflects the biomes. **b**, Same as **a** but for biomass density. **c**, Top 5
 215 countries with the highest contribution of non-forest tree biomass to the national biomass stocks, as well as distribution of
 216 biomass among land use/cover classes. **d**, Country-level biomass comparison between PlanetScope-based estimations and
 217 FAO statistics [5]. Forests are defined by GLULC and divided into broadleaf and coniferous using the Copernicus tree
 218 cover maps for countries in the European Economic Area (EEA) and the PROBA-V forest type map (2019) for countries
 219 outside the EEA (Belarus, Moldova, Russia, Ukraine) [32]. For more countries, see Extended Data Fig. 8.

Country	Total tree cover (% of country)	Tree cover outside forests (% of total)	Total biomass in Tg	Tree biomass outside forests in Tg (% of total)
Ireland	10.5%	19.7%	36.65	3.24 (8.8%)
UK	14.1%	22.1%	264.29	26.29 (9.9%)
Netherlands	14.8%	24.6%	47.32	5.79 (12.2%)
Denmark	16.9%	19.5%	63.47	4.74 (7.5%)
Ukraine	20.1%	11.4%	1534.2	53.2 (3.5%)
France	34.6%	9.7%	2388.2	77.2 (3.2%)
Germany	36.1%	10.0%	2264.3	65.4 (2.9%)
Italy	38.6%	7.5%	1501.3	37.6 (2.5%)
Estonia	50.3%	3.0%	259.3	2.7 (1.0%)
Finland	67.8%	1.7%	1726.2	9.76 (0.6%)

222
 223 **Table 1. Country statistics of tree cover and biomass.** Ten example countries sorted by the total tree cover percentage.
 224 Full country statistics are provided in Extended Data Table 1. The class trees outside forests was defined following the FAO
 225 definition.
 226

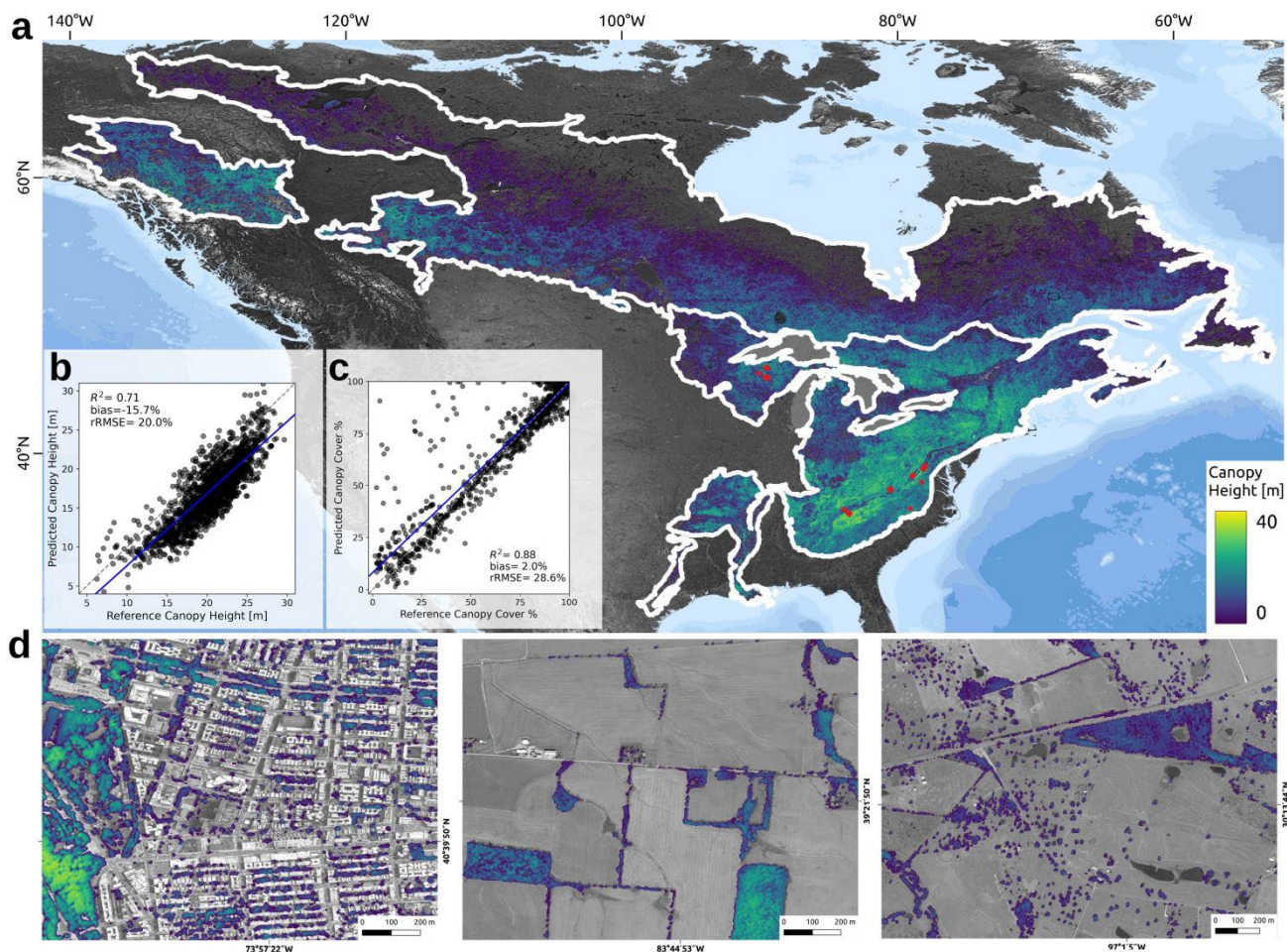


Figure 6. Canopy height estimation for the boreal and temperate zones in North America. **a**, Canopy height aggregated to 1-km resolution. **b**, Scatter-plot of canopy height from aerial CHM and PlanetScope predictions in test areas (red boxes in **a**) using canopy height averaged over 1x1 km sample sites ($n=1,036$). **c**, Same as **b** but for canopy cover percentage. **d**, Zoom-in examples of canopy height predictions in urban, cropland and grassland areas at 3 m resolution. The land basemap is from Google Maps satellite imagery (Imagery © 2022 CNES / Airbus, Landsat / Copernicus, Maxar Technologies, Map data ©2022). The ocean basemap in **a** is from www.naturalearth.com.

Beyond Europe

To test the scalability of our approach, we applied the tree cover segmentation and canopy height prediction to the boreal and temperate zones of North America and tested the performance with a fully independent LiDAR dataset (1,000 km² at 1 m resolution) that was never seen during training and parameter optimization (Fig. 6). Errors were found to be comparable to the European test datasets if aggregated to 1x1 km samples ($R^2=0.71$, bias=-15.7%, rRMSE=20% for height regression and $R^2=0.88$, bias=+2%, rRMSE=28.6% for tree cover segmentation). At 3x3 m, the overall RMSE was higher than in Europe (6.5 m). Note that a conversion to biomass would require region-specific calibrations with local field data, which is beyond the scope of this study.

Discussion

The PlanetScope-based canopy cover, height and biomass products provide a level of detail that was previously only possible with airborne data from LiDAR, which are costly and rarely available at national level. Airborne LiDAR-derived maps of woody resources are not available at continental scale, and spaceborne LiDAR from GEDI [34] is only available at a point scale and not over multiple years. Furthermore, GEDI and also the soon to be launched BIOMASS mission do not cover the northern parts of the world [35]. Previous studies combined sparse GEDI data with Landsat [24] or Sentinel-2 [26] to generate global scale canopy height maps, and although these perform well in forests, they have a large bias in non-forest areas (Fig. 2c,d). Moreover, the major strength of deep learning with 3 m resolution nanosatellite imagery lies in the fact that the model can learn features from the visible tree crown structure, which is often less clear in Sentinel-2 (10 m) or Landsat (30 m) images. In PlanetScope images at 3 m resolution, tree crowns and their shadows are visible representing key features when training deep learning methods to retrieve canopy heights.

The PlanetScope-based maps could be operationally integrated in national carbon stock inventory schemes. First, the spatial resolution allows to identify individual tree crowns, and makes it possible to align the satellite image with field plot data, which can be useful to reduce variances when upscaling plot data on e.g. carbon stocks to the national scale. Using different canopy height-to-biomass conversions for different forest and non-forest types account for the bias caused by the forest understory, which is undetected by classic aerial surveys. Aggregating our biomass maps to national scales shows a good alignment with national NFI statistics (bias 7.7% overestimation for 30 countries) [5], even though NFI plot-level data were only available for Denmark. Second, while the imagery is not free of charge, it is cheaper than aerial imagery, making it feasible to be acquired annually at national scales. The daily global coverage facilitates cloud free mosaics of different periods of the year, and once a model is trained, it has potential to be applied each year on an operational basis without further need of LiDAR data. Our example of North America demonstrates the geographical robustness of our models over trained biomes, but canopy height estimations in dry and tropical areas require further work. Third, LiDAR and aerial imagery differ between countries with regard to spectral, spatial, and temporal resolution, making it difficult to create biomass maps that are consistent between countries and years at continental scale. Although this was not demonstrated in this study, PlanetScope-based maps have a certain consistency in space and time [36,37], allowing for large-scale and multi-year biomass assessments. Fourth, integrating trees outside forests in national inventories is not only interesting from a carbon stock perspective, but also allows to quantify a variety of ecosystem services provided by these trees. A previous study used sub-meter LiDAR and aerial images to quantify the tree cover outside forests in Denmark, yielding the same proportion as this study (20%) [38], confirming the reliability of the PlanetScope-based map for the mapping of non-forest trees. For the UK, we found the fraction of urban trees outside forest cover to be 20.2%, which is in general agreement with the 16.5% reported in the UK NFI [17].

At continental scale, our study shows that trees outside forests do not play a critical role for the national above-ground carbon stocks of most countries of Europe. There are exceptions, however, and several countries have a tree cover outside forests that exceeds 20% of the national tree cover (Fig. 3b). Moreover, trees outside forests are not only valuable for their carbon stock, and a systematic identification of their location could be an integral part of annual monitoring and planning schemes related to biodiversity, microclimate, habitats, landscape values, and hydrological cycles [11,12]. Furthermore, the systematic integration of trees outside forests in tree cover maps could potentially improve climate and ecosystem modeling.

A drawback of the presented method is the requirement of large amounts of airborne LiDAR for training and validation, as well as NFI data for converting canopy cover and height into biomass, which currently limits the applicability primarily to the Global North. Although our map has shown robustness across Europe, the generalization to North America yields comparable results when aggregated at 1 km resolution, but performance at the 3 m pixel-level drops. Nevertheless, fine-tuning on some reference data from a new region of interest will likely close the performance gap. The conversion of height and cover into biomass can potentially be improved by more region-specific field data. In semiarid regions, small trees form a complex matrix of low vegetation, representing more challenging conditions for accurate predictions of the deep learning-based model. We confirm this with higher error values in the Mediterranean zone (Fig. 2a; Extended Data Fig. 3), where our map should be used with caution. Acknowledging these current shortcomings, recent development of state-of-the-art methods from the field of machine learning combined with cost-efficient nanosatellite imagery with a spatial resolution below 5 m open a new research avenue towards improved monitoring of national tree resources, both in and outside forests.

Methods

Summary

We used canopy height models (CHM) from airborne LiDAR data across Europe to train two deep learning models, one segmentation model predicting binary canopy/no-canopy (for trees taller 3 m height), and a regression model predicting canopy height, both from PlanetScope optical imagery at 3 m resolution. The canopy map was used as a mask and canopy height was only predicted within areas identified as tree cover. We included a GEDI and Sentinel-2 based canopy height map from ref [26] as an additional model input, both in the training and prediction process, providing a priori knowledge on the height distribution of forest area. We observed that this improved the underestimation of tall forest trees, and that in case of temporal or spatial mismatches (e.g. clearcuts and trees outside forests) the prediction followed the PlanetScope image overruling the information from the GEDI-Sentinel map. Here it is important to mention that the map from ref [26] severely overestimates trees in sparsely vegetated areas, which did however not impact on our result because we applied a binary canopy map at 3-m as mask (Extended Data Fig. 4). We used Danish NFI data to establish allometric relationships between plot based above-ground biomass and height (>1.3 m) from aerial CHMs averaged over the field plot. The relationship between canopy height and biomass was then used to convert canopy height and cover predicted from PlanetScope imagery into biomass at 30 m resolution (same as NFI plot), which was finally aggregated to the hectare level. We did not use the field measured height but the average vegetation height from the LiDAR data per plot because it comes closest to our predicted height. We also did not use PlanetScope predicted height because of alignment uncertainties with the field plots at 3-m resolution. The lack of publicly available GPS-referenced field plots covering the entire continent makes it impossible to train a model that can predict biomass directly from PlanetScope imagery.

Data

PlanetScope Imagery. We used PlanetScope imagery from Planet Labs available via a research license. The images are available daily for the entire globe with approximately 3 m pixel resolution and four multispectral bands: red, green, blue and near infrared. We generated yearly 1x1 degree tiles of mosaics from around 100,000 image scenes for the study area from 2019. Each mosaic tile consists of roughly 100 high quality cloud free scenes and were selected within a time window according to the MODIS phenology product [39]. The selected time window started 25 days before ‘senescence’ and ends 10 days before ‘midgreendown’, which reflects late summertime in Europe, where all woody vegetation still has green leaves and visible crown structure. For areas with frequent cloud cover, the time window was progressively extended towards earlier dates until the whole tile was fully covered with scenes. To reduce differences resulting from the mosaicking scenes collected from different sensors and times, we applied a histogram matching to Landsat and Sentinel-2 surface reflectance images from the same date range to produce homogeneous mosaics [37].

Airborne LiDAR canopy height maps. We collected publicly available airborne LiDAR CHM data in Europe and the USA, with full country coverage for Denmark (0.4 m resolution merged from 2016 to 2021, in total 43,000 km²), Estonia (1 m resolution from 2011 to 2017, in total 45,000 km²), the Netherlands (0.2 m resolution from 2018, in total 41,000 km²) and Spain (2.5 m resolution from 2012, in total 500,000 km²). Other samples were available for patches in Finland (1 m resolution from 2019, in total 50,000 km²), Switzerland (0.5 m resolution merged from 2017 to 2020, in total 20,000 km²) and Wales (1 m resolution from 2015, in total 10,000 km²) (Extended Data Fig. 1b). Some data were provided as CHM, others as raw point cloud points, which were converted to DTM and DSM used to calculate CHMs via LAStools [40]. CHM data represents the top canopy height in the pixel area, but sometimes includes the height from buildings and short herbaceous and bush vegetation. Here we used the building footprints from Microsoft [41] and Open Street Map [42] to mask buildings. We further visually compared PlanetScope imagery and the corresponding CHM, and decided for a 3 m height threshold to separate tree crowns from shrubs, bushes and grasses. Woody plants below this threshold were not clearly visible in the PlanetScope images.

Training samples generation. To adjust airborne CHM with PlanetScope images, we reprojected the PlanetScope imagery to match the aerial data UTM projection, and resampled the aerial CHM to the spatial resolution of PlanetScope data (3 m) by reserving maximum values (filling small gaps in forests in the aerial CHM which are not visible in PlanetScope imagery). We applied a weighted sampling strategy to the aerial CHM data, according to collection year, mean height, tree cover percentage (area > 3 m), forest type (coniferous or broadleaf [32]) and dominating land use/cover classes, to control the training data distribution fitting into the model and to minimize possible mismatches between data sources caused by acquisition time differences (Supplementary Fig. 2). The final dataset contains 100,000 samples each 328x328 pixels corresponding to 1x1 km ground spacing, where 10% were used for validation, 10% as test dataset, and 80% for training the model.

Mapping tree cover and height using deep learning

We used the U-Net architecture [43] with a EfficientNet-B4 backbone [44] for both the tree cover segmentation and the tree height regression tasks (Supplementary Fig. 1). The modified U-Net has proven to be able to delineate tree canopies in very high resolution imagery [11,37,45]. EfficientNets have shown to perform well with small parameter size by a pre-designed

network depth, width, and resolution. Since we need to deploy our model to continental scale at 3 m resolution, we chose the EfficientNet-B4 as the trade-offs between performance and predicting (training) time. A key objective of our work was to assess tree resources outside forests as accurately as possible, and the crown size of such trees is often just a few pixels in our PlanetScope imagery. In the decoder, we replaced the simple two convolutional layers in the U-Net by residual blocks [46], which has been demonstrated to improve the model performance for segmenting small objects [47]. Furthermore, spatial and channel squeeze and excitation (scSE) blocks were integrated in the decoder to build attention mechanisms [48]. The deep learning framework is built on an open-source segmentation library [49].

Mismatches between aerial CHM and PlanetScope data were sometimes observed in forested areas, when the images were not acquired from the same date. These inconsistencies cumulatively contribute to a significant proportion of loss in the regression, resulting in an unstable training progress and large uncertainties in the tree height estimations. Therefore, we first segmented the tree canopies, which were then used to mask the non-forested areas for minimizing the regression loss (Supplementary Fig. 1).

Tree crown segmentation. We set a 3 m height threshold to the aerial CHM data and converted it into a reference binary (tree or background) map. The 4 band input data from PlanetScope were augmented by random cropping, flipping, brightness distortion and then normalized channel-wise based on training dataset statistics before feeding into the model. We used the focal Tversky loss [50] as loss function for the segmentation model, the alpha and beta parameters in the Tversky index [51] penalising false positives and false negatives were set to 0.4 and 0.6. The focal parameter was set to 1.2, which incentivises the model to focus on harder examples (when the Tversky index < 0.5), especially for the cases where scattered trees or tree lines do not exactly overlap between PlanetScope and aerial CHM data. We used Adam [52] as optimizer with an initial learning rate of 0.0001, then reduced the learning rate by a factor of 0.1 when the metric has stopped improving for 30 epochs. Model training stopped when the learning rate reached $1e-7$. The model was then used to predict a binary tree/non-tree cover map at 3-m for the entire study area.

Tree height regression. We used the same network architecture and training strategy as for the segmentation task, with an inverse sample-frequency weighting smooth-L1 loss function to mitigate the underestimation that is often observed for tall trees [26]. The regression loss was only calculated for areas that overlap between the segmentation tree mask and the aerial CHM (> 3 m). While this largely eliminated uncertainties from mismatches between the two data sources, the model still underestimated taller forests, possibly resulting from unclear training data. For example, coniferous forests vary optically only little in PlanetScope image, but can have different heights in the CHM, depending on their location. As a consequence, whenever the model encounters this situation, it conservatively predicts intermediate values. One solution is to encode geo-information into the model [47] or include as input data [26]. However, this would require a more homogeneous distribution of the reference data. We thus included a GEDI/Sentinel-2 based height map from ref. [26] as auxiliary data (additional input band) to provide prior knowledge to the model about the forested areas, resulting in a total of 5 bands for the tree height regression task.

Allometry for biomass estimation

Allometric equations provide above-ground biomass estimates from related tree measurements, such as stem diameter at breast height, canopy height, or crown diameter [54]. We associated plot level biomass derived from species specific equations and field measured stem diameters with airborne LiDAR CHM (> 1.3 m) averaged over the same plots [54,55]. The plot based above-ground biomass includes trees of all sizes above 1.3 m including understory, and relating the total above-ground biomass value per plot with the average canopy height derived from airborne LiDAR (for all areas with woody vegetation > 1.3 m, following the NFI definition) automatically corrects for the bias from undetectable understory. The regression equation in the log-log space can be described as:

$$\ln(\text{AGBD}) = \alpha + \beta * \ln(H) + \varepsilon \quad (1)$$

where AGBD is the above-ground biomass density (in kg/m^2), H is the average canopy height for woody vegetation taller than 1.3 m for the NFI plot area from the aerial CHM, α and β are the regression coefficients, ε is normally distributed error term with zero mean and standard deviation σ , that is, $\varepsilon \sim N(0, \sigma^2)$. When back-transforming model (1), we corrected for logarithmic bias using the Baskerville correction [56].

Plot level biomass data were from the Danish NFI ($n = 11,296$) collected between 2012-2021 [33]. Each plot was a circle with a radius of 15 m, where averaged tree height, estimated above-ground biomass, and forest fraction were provided. To remove misaligned plots, we filtered plots where the forest cover difference was more than 30% but kept data regardless of the collection year, assuming that dynamics are averaged out by the large number of plots. We derived forest type specific relationships using the Copernicus forest map from 2018 [32] to classify the plots in broadleaved forest (bf), coniferous forest (cf), mixed forest (mf) and areas of sparse tree cover (nf), which are plots that fall outside of the areas mapped as forest in the Copernicus map (Extended Data Fig. 6). We acknowledge that the class representing the non-forest trees (nf) is not ideal, because the Danish NFI does not systematically measure trees outside forests and other wooded lands. However, here we assume that plots being located in areas of sparse tree cover and having a low biomass are most representative for non-forest areas. Since the NFI biomass data includes all trees taller 1.3 m and not only those visible from an aerial

perspective and in the PlanetScope images (typically >3 m height and 10 m² crown size), the different relationships include understory or small trees, which indirectly corrects for the bias of missing small trees in the PlanetScope based tree height predictions. This fact makes it important to have a separated equation for trees outside forests, which typically have a lower amount of understory. The equations and the number of NFI plots are as follows:

$$AGBD_{bf} = \exp[-2.427] * H^{1.794} * \exp[0.134^2/2] \quad (n_{\text{plots}} = 7232) \quad (2)$$

$$AGBD_{cf} = \exp[-1.652] * H^{1.547} * \exp[0.152^2/2] \quad (n_{\text{plots}} = 3768) \quad (3)$$

$$AGBD_{mf} = \exp[-2.243] * H^{1.728} * \exp[0.118^2/2] \quad (n_{\text{plots}} = 7536) \quad (4)$$

$$AGBD_{nf} = \exp[-2.227] * H^{1.742} * \exp[0.305^2/2] \quad (n_{\text{plots}} = 409) \quad (5)$$

We used 20% of the plots randomly selected for each type to derive error metrics shown in Extended Data Fig. 6. Overall, the equations showed a reasonable performance with a relative low bias (averaged R²=0.62; bias=-8.7%; RMSE=5.43 kg/m²).

Plot-level above-ground biomass in kg was then estimated as:

$$AGB = (TCP / 100) * 900 * AGBD \quad (6)$$

where TCP is predicted tree cover percentage aggregated in 30x30 m resolution (rasterized NFI plot size). AGBD is obtained from equation 2-5 according to the dominant forest type within the plot area. The above-ground biomass at 30 m resolution was finally aggregated to the hectare level, containing both forest and non-forest information (Fig. 4).

Evaluation

We randomly selected 10% of the CHM dataset (10,000 1x1 km samples) as independent test dataset, which was not used for training and to optimize the model parameters during training, neither for the tree cover segmentation nor for the tree height regression tasks. We evaluated our results at pixel level (3x3 m) using the intersection over union (IoU) for the segmentation, the root mean squared error (RMSE) for tree height regression; and aggregated to 1x1 km grids using the relative systematic error (bias %, equation 7), positive bias represents an overall overestimation and negative bias represents an overall underestimation [54]:

$$\text{bias} = \frac{1}{N} \sum_{i=1}^N \left(\frac{Y_{\text{pred}} - Y_{\text{true}}}{Y_{\text{true}}} \right) \times 100 \quad (7)$$

and the relative RMSE (rRMSE) in percentage described in equation 8:

$$\text{rRMSE} = \sqrt{\frac{1}{N} \sum_{i=1}^N \left(\frac{Y_{\text{true}} - Y_{\text{pred}}}{Y_{\text{true}}} \right)^2} \times 100 \quad (8)$$

Tree cover segmentation: Overall, our model achieved an IoU of 0.692 on the test dataset (n = 10,000 plots of 1x1 km). Further evaluation revealed a good alignment between our tree cover map and the tree cover map from aerial CHM data (using a 3 m height threshold) with overall R² values of 0.92 but with variation across biomes (Extended Data Fig. 2e). Trees can be segmented more precisely in both boreal (R²=0.97; bias=-1.0%; rRMSE=13.5%) and temperate zones (R²=0.99; bias=-0.8%; rRMSE=18.5%), than in the Mediterranean zone (R²=0.39; bias=-8.2%; rRMSE=89.8%), which is likely due to two reasons: First, a limited amount of training data was usable from Spain because of the large temporal gap between PlanetScope imagery and the available aerial CHM data, which is from 2012. Second, the method of using a constant height threshold to generate the tree mask is not fully functional in diverse vegetation structures consisting of trees and shrubs. For areas with trees outside forest, our predictions compared with aerial CHM data achieve an overall bias=-2.4% and rRMSE=36.6%, (Extended Data Fig. 3; Fig. 2d). We also calculated a confusion matrix comparing Danish NFI plots (n = 13,638) classified as either forest or no forest with our data following the FAO definition, showing very high agreement with an overall accuracy of 0.96 (Extended Data Fig. 7a).

Tree height regression: We evaluated our tree height model by a residual analysis between reference height from the test dataset and the corresponding predicted tree height at pixel level (3m resolution) (Fig. 2a). The model performance varied across biomes and canopy height ranges, yet was closely related to the data distribution (Fig. 2b). In general, the model did not show a considerable saturation for taller canopies, yielding a RMSE of 5.4 m and a MAE of 4.2 m. We found similar metrics as for the segmentation across biomes at plot level evaluation, with R²=0.82, bias=-2.4%, rRMSE=17.9% for the boreal zone, R²=0.9, bias=-1.6%, rRMSE=14.5% for the temperate zone and R²=0.45, bias=-20.1%, rRMSE=35.4% for the Mediterranean zone (Extended Data Fig. 2d). All slopes below the diagonal line indicated a slight underestimation, consistent with the pixel-level evaluation (Fig. 2a). As for trees outside forests, we obtained a bias of -1.6% and rRMSE of 26.4% (Extended Data Fig. 3a). We further evaluated our top height estimation with field measured plot scale height from Danish NFI data (n = 3,451) (Extended Data Fig. 7b), showing an underestimation bias of 9%.

North America. To test the generalization of our model, we then predicted tree cover and height for the boreal and temperate zone of North America, and compared the results to aerial CHMs which were not used for either training or validation of the model parameters. Here we acquired around 1,000 km² of airborne LiDAR CHMs at 1 m resolution from ref [53], and generated 1,036 samples with 1x1 km² ground spacing following the same data processing procedure as described in the previous section. This evaluations show that our model can be deployed to a comparable landscape without a considerable loss of performance ($R^2=0.71$, bias=-15.7%, rRMSE=20% for height regression and $R^2=0.88$, bias=+2%, rRMSE=28.6% for tree segmentation) (Fig. 6).

Biomass estimations: The errors from canopy cover, height and the allometric equations propagate to the final biomass estimation, so we evaluated the uncertainty on the biomass maps in three ways. First, we compared the predicted biomass at field plot level with estimations from the Danish NFI for 3,451 selected plots, resulting in an overall bias of -23% (Extended Data Fig. 8c). Since NFI data were collected in forest areas, this error is impacted by the underestimation of very tall trees, where biomass densities can reach 300 Mg/ha (Extended Data Fig. 8c) and possible geolocation uncertainties between the PlanetScope images and the NFI plot locations recorded by a handheld GPS. Second, we compared our biomass estimation with other global biomass products [28,31,27]. The GEDI biomass product [31] does not cover our entire research area, so we randomly sampled 4,000 points (Extended Data Fig. 7d,e,f) resulting in a bias of 1.8%. The biomass map from reference [28] shows good correlation with ours but with a higher bias (bias=11.3%, $r=0.83$). The biomass map from reference [27] shows an overall underestimation (bias=-37.9%, $r=0.82$). Third, we further compared the PlanetScope-based biomass maps aggregated to national scale with FAO country statistics (Fig. 5d, Extended Data Fig. 8a) [5]: The overall bias over all 30 countries was 7.6%. Our biomass estimates reached similar levels but were slightly higher than the FAO statistics for most countries, which could result from trees outside forests that are typically not included in national inventory reports. The total bias for ref [28] and ref[15] at country scale was calculated as +17.3% and +49.9% .

Environmental and auxiliary data

We used a Global Land Cover and Land Use map from 2019 at 30 m [29] including forests, urban areas, cropland and grassland to analyse trees outside forests and trees in different landscapes. We also used global canopy height products based on Sentinel-2 from 2020 at 10 m resolution [26] and another product based on Landsat from 2019 at 30 m resolution [24] for comparison. Both products use GEDI spaceborne canopy height data which was extrapolated with Sentinel-2 and Landsat, respectively. The GEDI/Sentinel-2 based canopy height map was also used to assist the height regression task. The Copernicus forest type map based on Sentinel-2 data from 2018 at 10 m resolution and a PROBA-V based forest type map from 2019 at 100 m [32] were used to separate forests into coniferous and broadleaved forest across Europe. We further used the GEDI L4B biomass density product from 2020 available at 1 km resolution [31] and state-of-the-art global biomass density products from 2017 [28] and 2020 [27] at 100 m for biomass comparisons. We used biome data from [30] aggregated into three main biome zones (Extended Data Fig 2).

Data and code availability

PlanetScope imagery was purchased via a departmental license and the images cannot be distributed. Derived products produced in this study: canopy height, cover and biomass maps across Europe will be made available upon acceptance of the manuscript. Links to download the open LiDAR data can be found under

https://publications.jrc.ec.europa.eu/repository/bitstream/JRC126223/jrc126223_jrc126223_lidaropensourcedata.pdf.

The code for tree segmentation and height regression is built on publicly open-source framework Segmentation Models PyTorch, available at https://github.com/qubvel/segmentation_models.pytorch. Custom code will be made available upon acceptance of the manuscript.

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Author contributions

M.B., R.F. and S.Liu designed the study. M.B. prepared and processed the aerial LiDAR CHM data. T.N. provided Danish national forest inventory data. F.R. and X.T. developed the code for the PlanetScope imagery generation pipeline. J.C. designed the biomass allometry, implemented by S.Liu, N.L., C.I., S.Li, M.M.. S.Liu wrote the code for the deep-learning framework, supported by F.R., S.Li and Z.C.. Interpretations were done by M.B., S.Liu, N.L., P.C., J.C., C.I., Y.Y., R.F. and S.Liu conducted the analyses. S.Liu and M.B wrote the first manuscript draft with contributions by all authors. S.Liu designed the figures.

551 **Competing interest**

552 The authors declare no competing interests.

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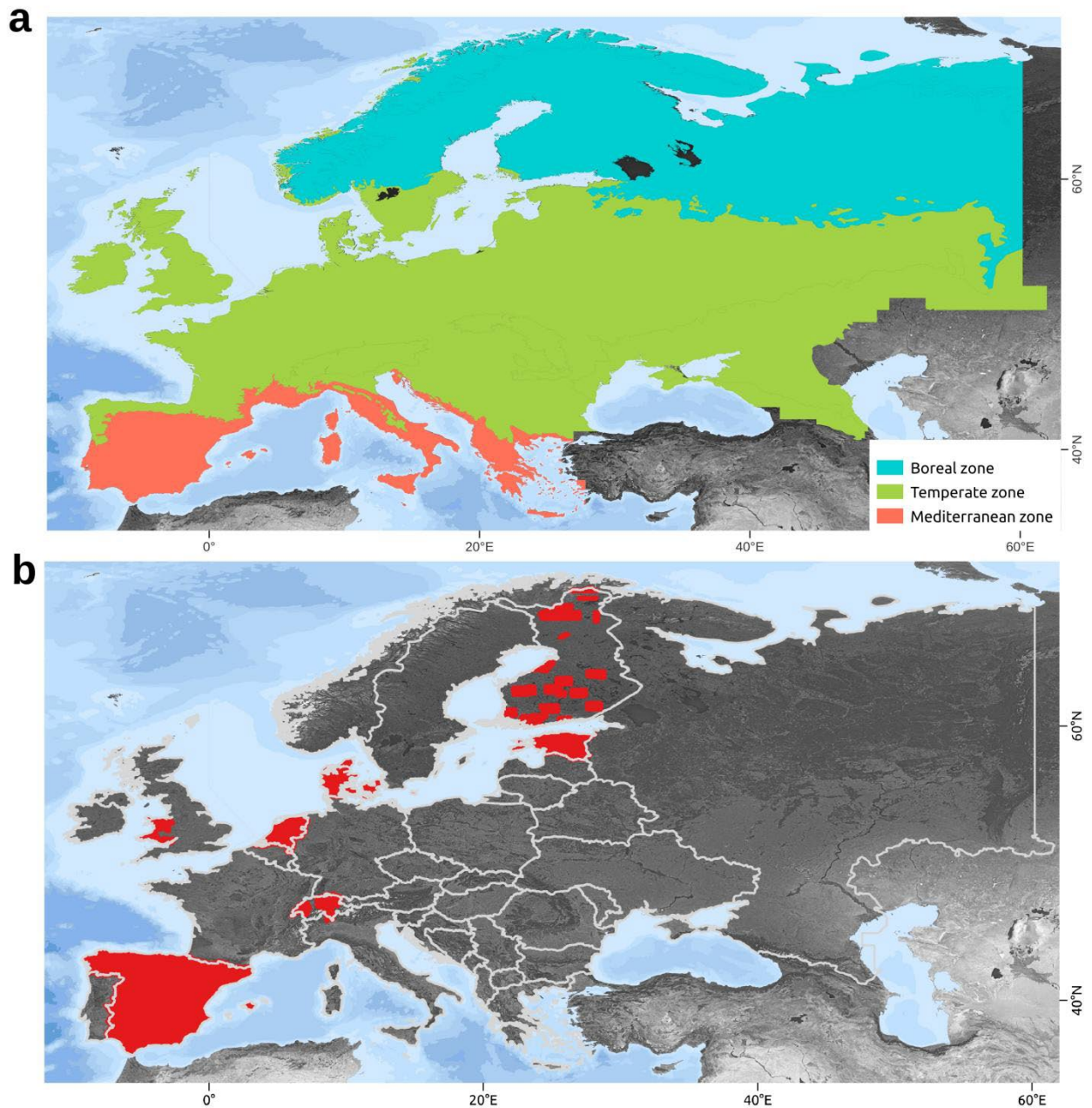
554 **Materials & Correspondence**

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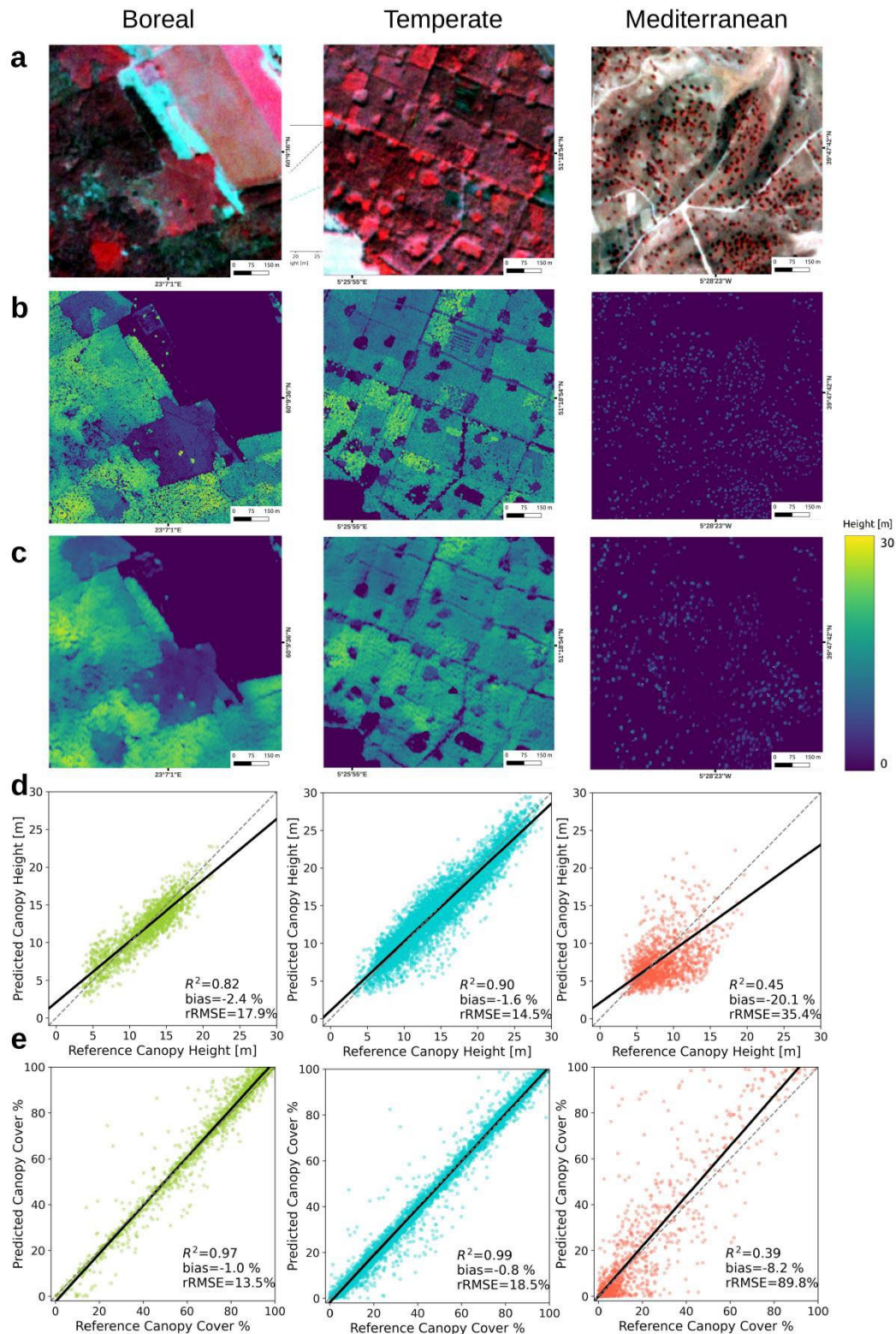
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Extended Data Fig. 1: Research area and datasets used in this study. a, Biomes are aggregated into biomes from data provided in reference [30]: The boreal zone includes “Tundra” and “Boreal Forests/Taiga”; the temperate zone includes “Temperate Broadleaf & Mixed Forests”, “Temperate Conifer Forests” and “Temperate Grasslands, Savannas & Shrublands” The Mediterranean zone is the “Mediterranean Forests, Woodlands & Scrub” biome. **b,** Location of airborne LiDAR CHM data. The land and ocean basemaps are from Google earth map and www.naturalearth.com.



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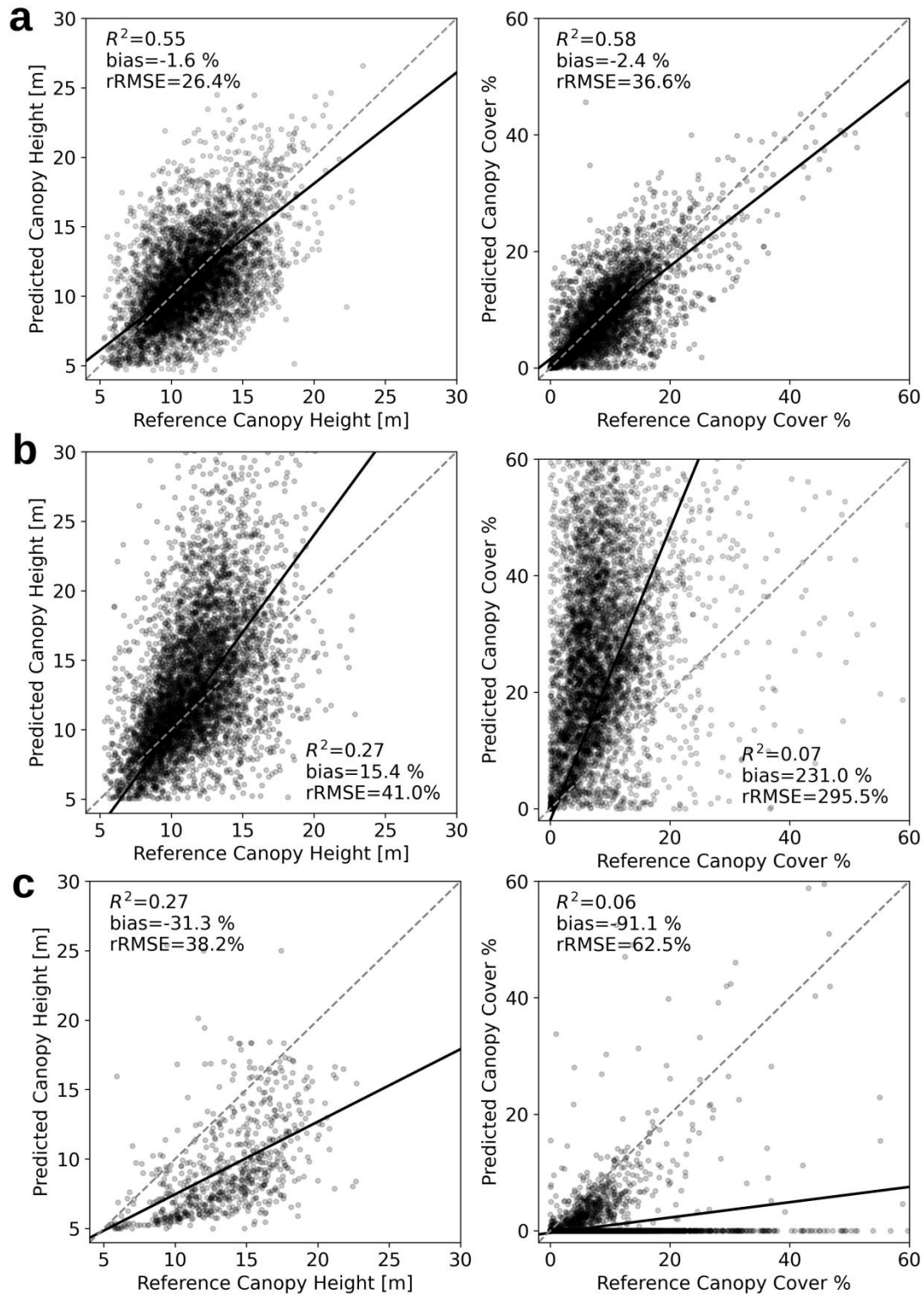
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Extended Data Fig. 2: Evaluation using airborne LiDAR grouped by biomes. **a**, PlanetScope imagery displayed in near infrared, red and green false color composites. **b**, Corresponding canopy height reference from airborne LiDAR. **c**, Corresponding canopy height prediction from **a**. **d**, Scatter plots between canopy height from airborne LiDAR and PlanetScope predictions. **e**, Same as **d** but for canopy cover. Number of 1x1 km samples for Boreal, Temperate and Mediterranean biomes: 1885, 7136, 979.



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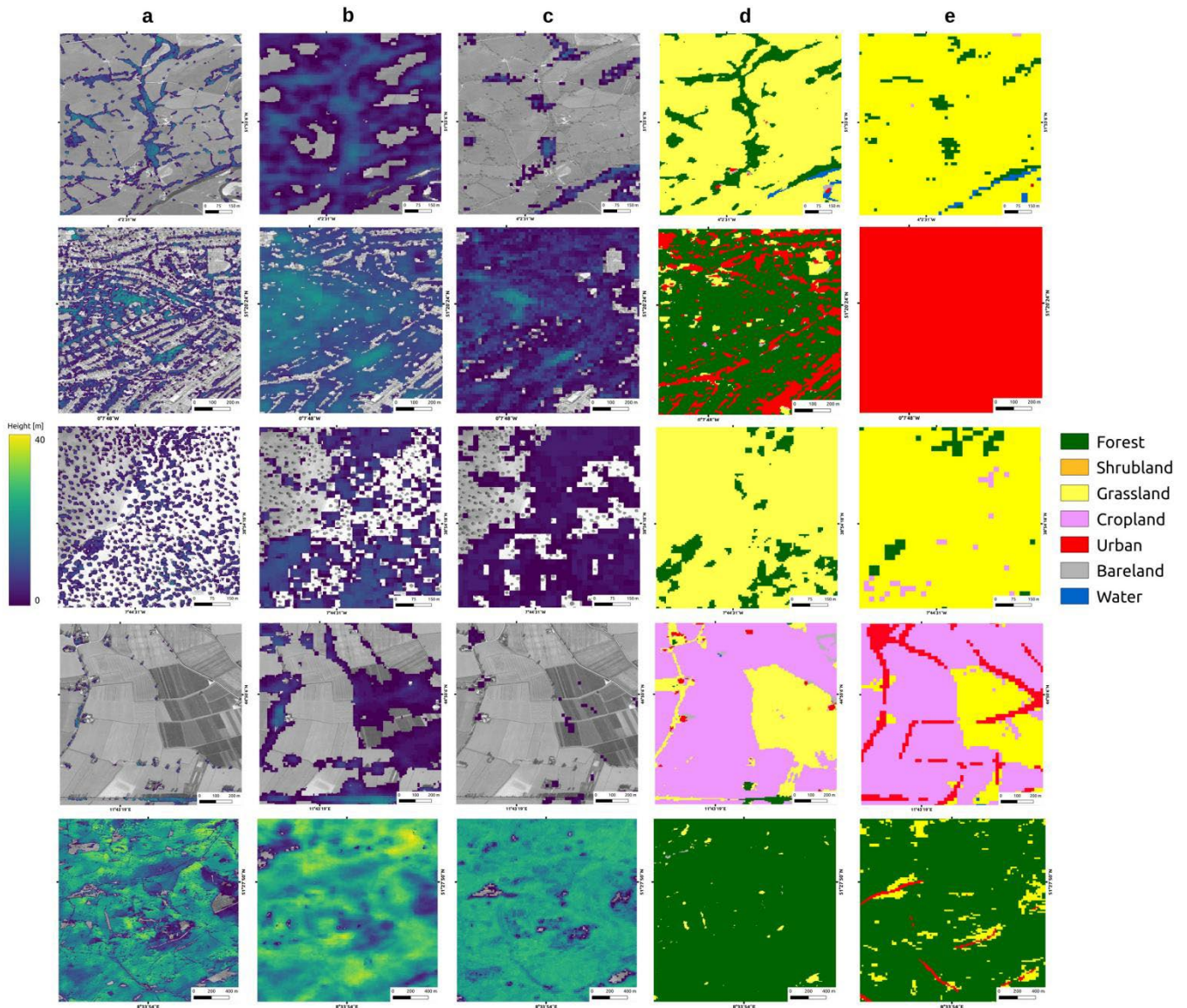
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Extended Data Fig. 3: Evaluation using airborne LiDAR for tree cover outside forests. **a**, Comparison of PlanetScope-based predictions for canopy height (left) and canopy cover (right) with aerial CHMs for trees outside forest, here defined by the global land cover land use dataset from ref [29]. **b**, Same as ‘a’ but for a Sentinel-2-based height product from ref [26]. **c**, Same as ‘a’ and ‘b’ but for a Landsat-based canopy height product from reference [24]. Samples from the Landsat-based product were only acquired in areas covered by GEDI data. Number of 1x1 km samples = 2,420.



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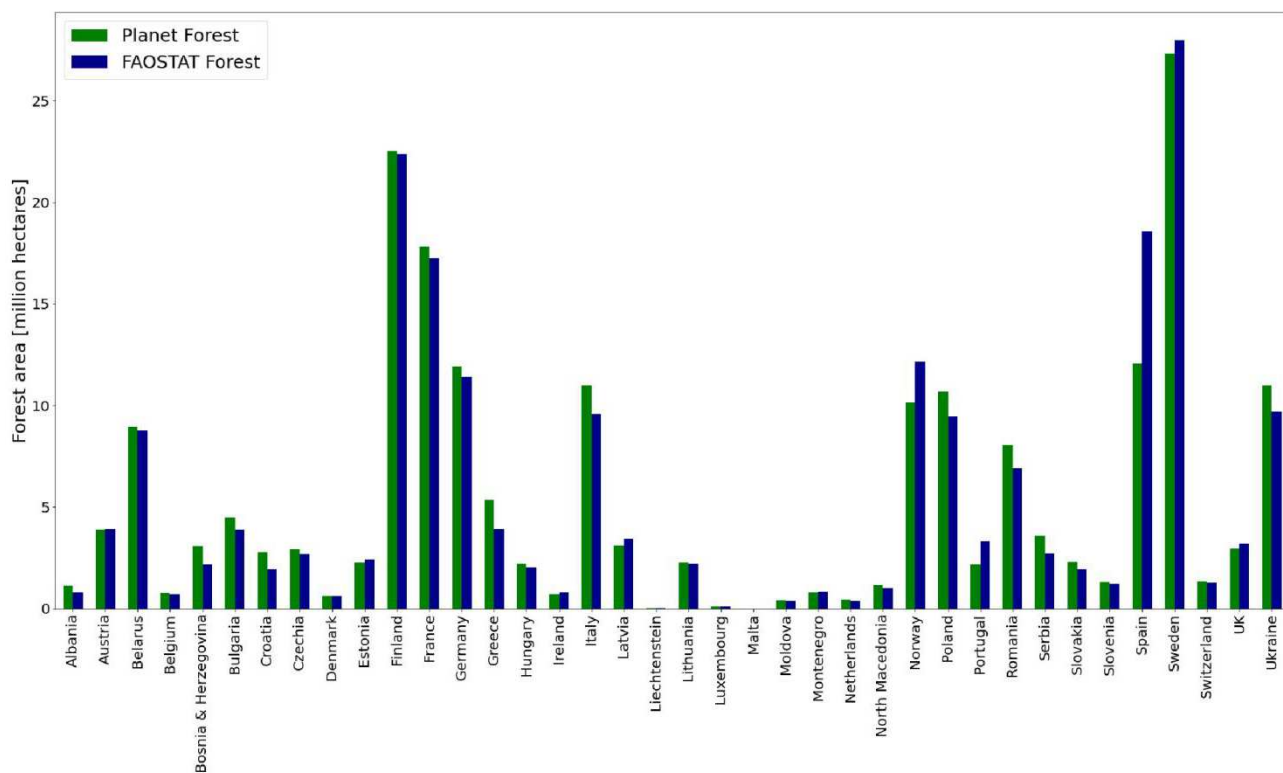
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Extended Data Fig. 4: Visualisation of satellite-based canopy height and land use/cover products. **a**, PlanetScope canopy height prediction from this study. **b**, Sentinel-2-based canopy height product from ref [26]. **c**, Landsat-based canopy height product from ref [24]. **d**, WorldCover land use/cover product [19]. **e**, GLULC land use/cover product from ref [29] that was used in this study. All canopy height images are displayed including trees above 5 m.

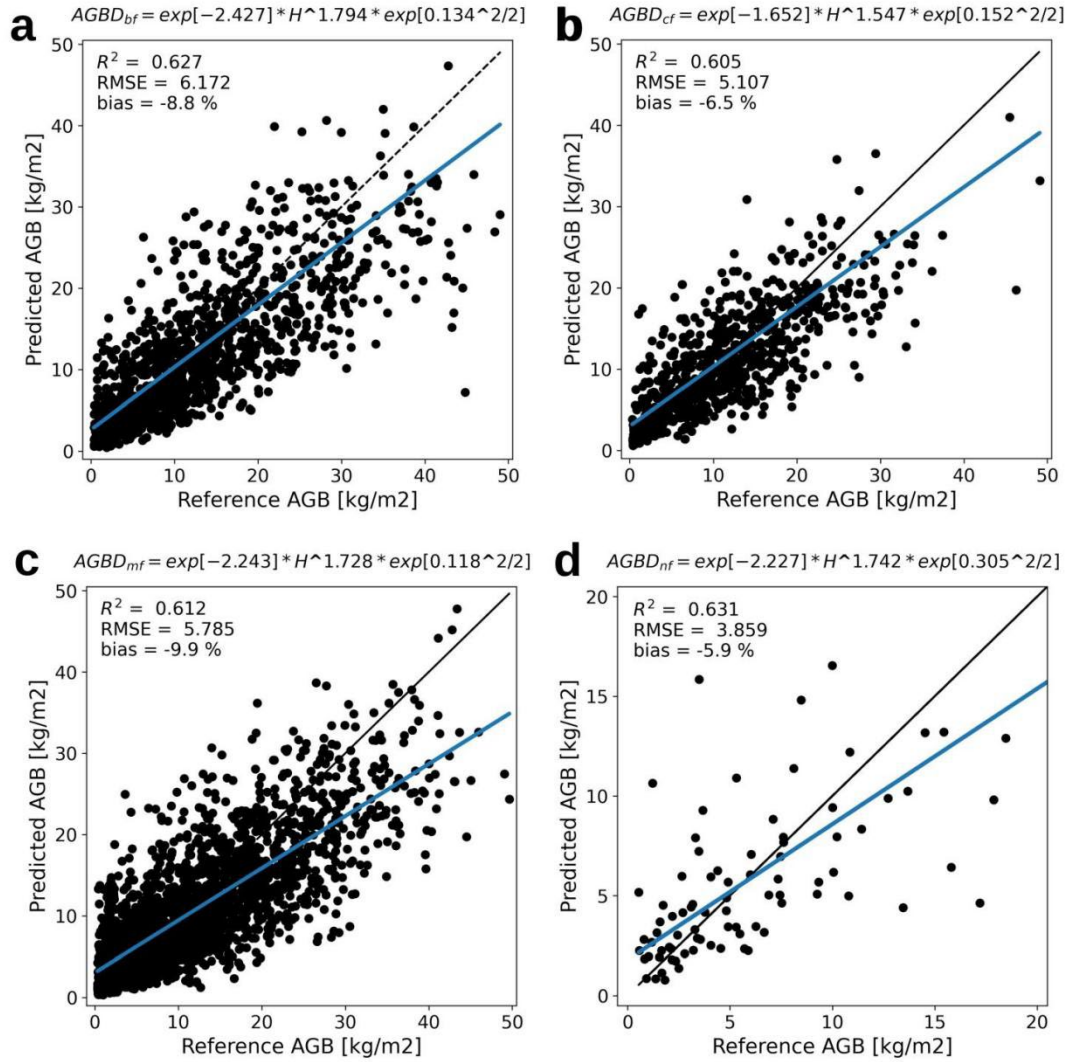


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Extended Data Fig. 5: Comparison with FAO statistics at country scale. Forest is defined by FAO as areas larger than 0.5 ha with trees taller than 5 meters and a canopy cover of more than 10 percent, excluding agricultural and urban land.



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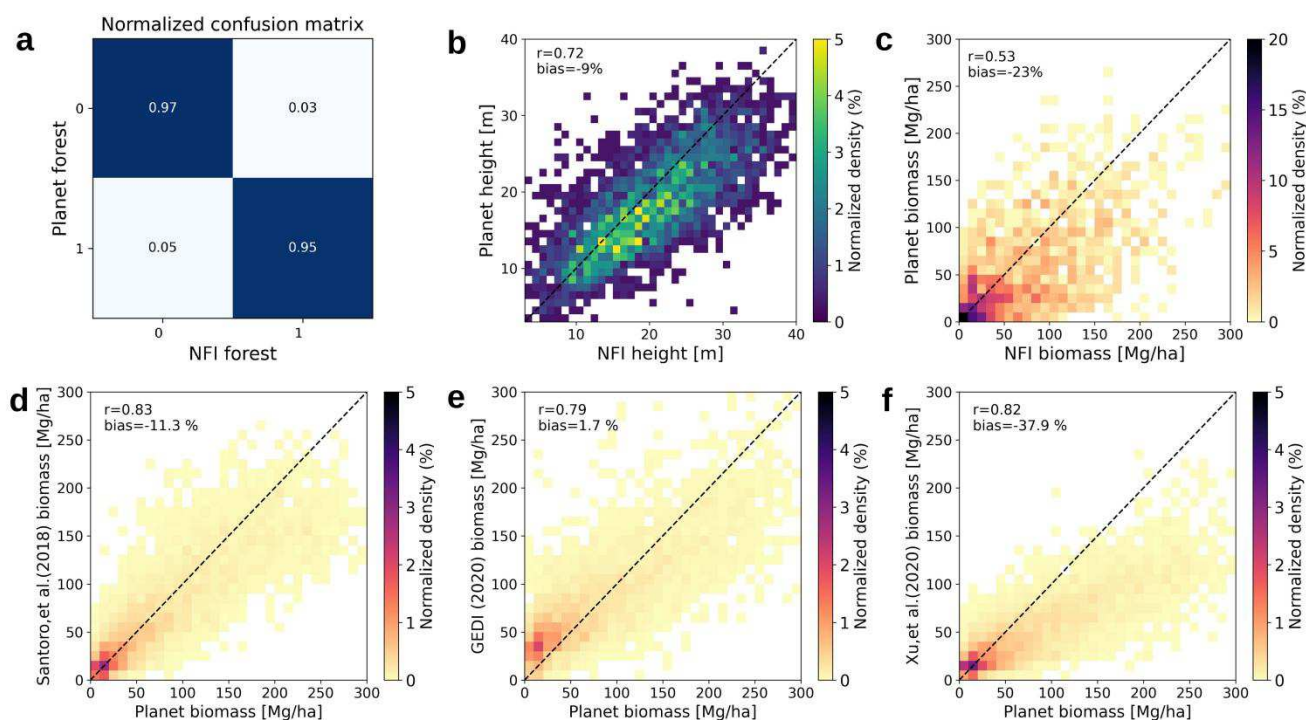
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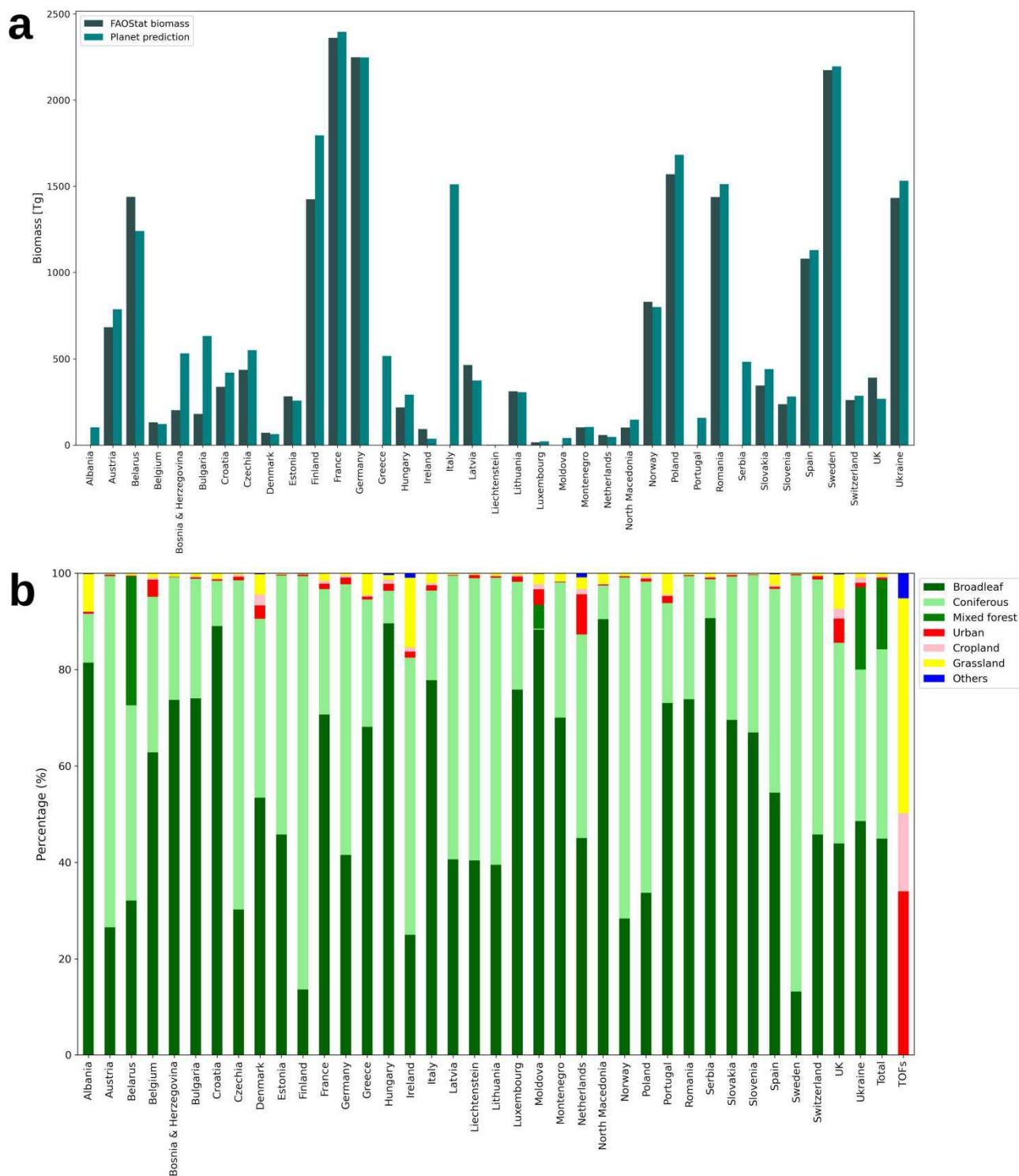
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Extended Data Fig. 6: Allometric equations for biomass conversions. **a-d**, Allometric equations derived from Danish NFI plot data and aerial CHM data for broadleaf forest ($n = 7,232$) (**a**), coniferous forests ($n = 3,768$), (**b**), mixed forests ($n = 7,536$), (**c**) and non-forest plots ($n = 409$) (**d**). The total number of NFI plots is 11296. AGBD is above-ground biomass density in kg/m^2 , H is the average height in meter.



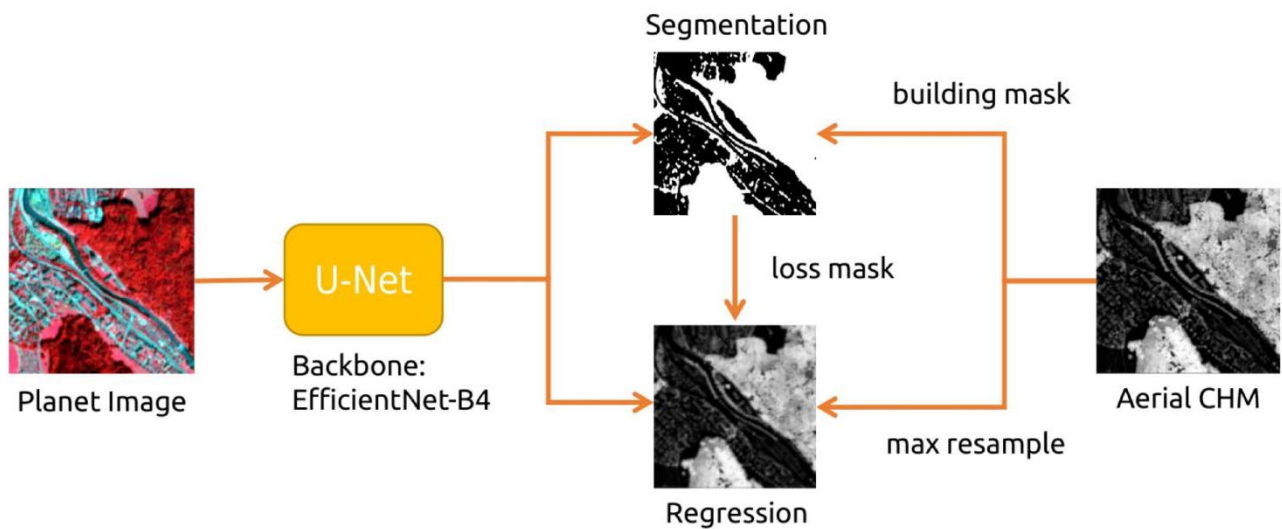
Extended Data Fig. 7: Evaluation using Danish NFI data [33] and comparison with other biomass products. a, Normalized confusion matrix comparing NFI plots (classified as forest or non-forest) with PlanetScope tree cover predictions. Here we followed the FAO forest definition using a 10% tree cover threshold combined with land use data to define if the area of the plot was predicted as forest or non-forest. **b,** Scatter plot between tree height recorded per NFI plot ($n = 3,451$) and our PlanetScope predicted top height. **c,** Same but for biomass. Number of samples: 13,638 collected between 2018-2020 for forest (3,451) and non-forest (10,187) plots. We applied a 3 m height threshold and 3 Mg/ha biomass density threshold to (**b**) and (**c**). **d,** Comparison of PlanetScope predicted biomass density with the global biomass map from ref [28]. **e,** Same but for the GEDI L4B biomass map in 2019 [Target not found!]. **f,** Same but for the global biomass map from ref [27]. All sources were aggregated to 1x1 km and a total of 4,000 samples were randomly selected. Height data was binned into 1 m width intervals, biomass density data was binned to 10 Mg/ha width intervals.



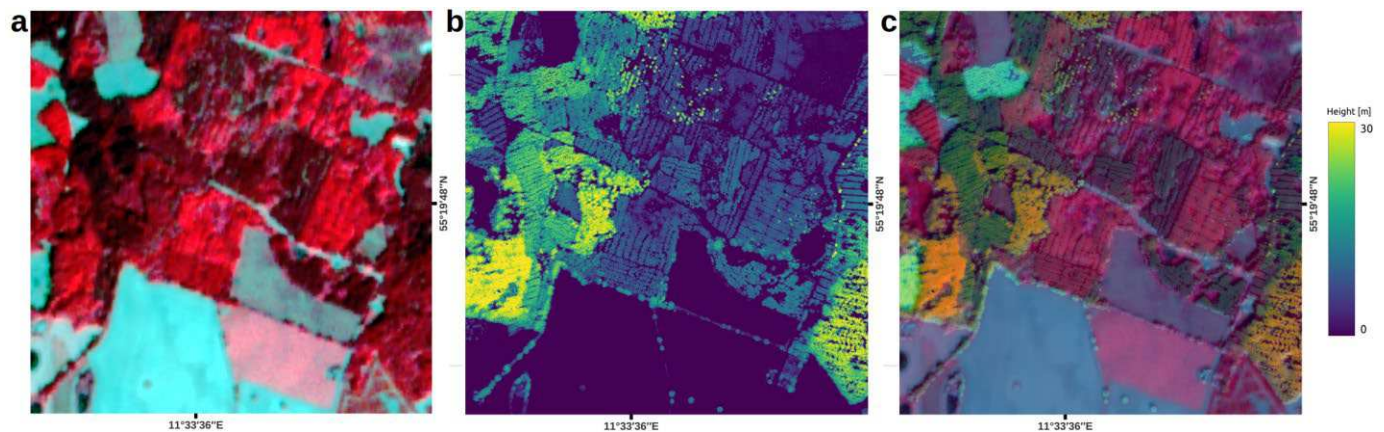
Extended Data Fig. 8: Biomass statistics at country scale. **a**, Comparison with FAO biomass at country scale. The FAO statistics are in carbon units [5], and our above-ground biomass maps aggregated at country scale were divided by 0.47, accordingly [58]. The bias between the PlanetScope based predictions and the FAO statistics is 7.6%. **b**, Biomass contributions of different forest and land use/cover types for each country. The underlying forest and land use/cover maps are from ref [32] and ref [29].

Country	Canopy Cover [1000 ha] / Biomass [Tg]							
	Forest & OWL		Trees Outside Forests					
			Urban		Cropland		Total	
Albania	1095.5	98.60	11.7 (1.0%)	0.34	3.9 (0.3%)	0.1	56.3 (4.9%)	1.73
Austria	3796.2	781.10	66 (1.7%)	2.95	16.4 (0.4%)	0.59	163.2 (4.1%)	7.45
Belarus	8799.9	1234.07	54.2 (0.6%)	2.02	54.5 (0.6%)	1.79	271.3 (3.0%)	11.81
Belgium	723.0	113.62	97.1 (11.1%)	4.58	12.6 (1.4%)	0.47	145.8 (16.8%)	6.97
Bosnia & Herzegovina	3023.7	476.90	15.3 (0.5%)	0.48	5.9 (0.2%)	0.17	83.5 (2.7%)	4.03
Bulgaria	4403.9	563.77	71.2 (1.5%)	2.26	53.2 (1.1%)	1.42	208.5 (4.5%)	7.44
Croatia	2710.4	416.56	36.6 (1.3%)	1.39	12.7 (0.4%)	0.48	114.4 (4.0%)	5.36
Czechia	2848.2	537.08	99.7 (3.2%)	4.45	51 (1.7%)	2.2	226.1 (7.4%)	11.56
Denmark	586.5	58.73	48.3 (6.5%)	1.75	44.4 (6.0%)	1.26	142.3 (19.5%)	4.74
Estonia	2214.6	256.62	12.9 (0.6%)	0.52	8.6 (0.4%)	0.27	67.6 (3.0%)	2.66
Finland	22535.0	1716.40	113.1 (0.5%)	6.68	31.1 (0.1%)	0.91	385.1 (1.7%)	9.76
France	17143.0	2311.02	702.3 (3.7%)	28.23	405 (2.1%)	14.28	1850.0 (9.7%)	77.21
Germany	11604.9	2198.99	720.8 (5.6%)	34.52	246.7 (1.9%)	9.27	1295.9 (10.0%)	65.35
Greece	5242.1	506.94	89.5 (1.6%)	3.03	50.3 (0.9%)	1.56	275.7 (5.0%)	9.32
Hungary	2148.1	279.36	119.4 (4.9%)	4.52	64.4 (2.7%)	2.42	258.3 (10.7%)	10.34
Ireland	590.9	33.41	13.1 (1.7%)	0.39	11 (1.4%)	0.25	144.7 (19.7%)	3.24
Italy	10732.3	1463.67	393.2 (3.4%)	16.57	174.2 (1.5%)	6.38	871.1 (7.5%)	37.6
Latvia	3037.1	370.27	16.9 (0.5%)	0.66	13 (0.4%)	0.36	106.9 (3.4%)	4.31
Liechtenstein	6.8	1.28	0.3 (4.1%)	0.01	0	0	0.5 (6.9%)	0.02
Lithuania	2187.0	303.62	25.9 (1.1%)	1.02	17.8 (0.8%)	0.52	103.1 (4.5%)	4.45
Luxembourg	93.7	20.87	5.2 (5.0%)	0.25	1.4 (1.3%)	0.05	9.4 (9.1%)	0.51
Malta	0.4	0.02	0.3 (27%)	0.01	0	0	0.6 (60.0%)	0.01
Moldova	390.5	38.08	49.9 (10.3%)	1.41	16.8 (3.5%)	0.42	87.2 (18.3%)	2.54
Montenegro	799.6	103.98	3.9 (0.5%)	0.12	0.3 (0.0%)	0.01	20.5 (2.5%)	0.87
Netherlands	418.5	41.53	84.9 (15.1%)	3.92	11.8 (2.1%)	0.46	136.2 (24.6%)	5.79
North Macedonia	1114.7	143.68	8.9 (0.8%)	0.3	6.9 (0.6%)	0.19	44.2 (3.8%)	1.79
Norway	10133.0	821.10	48.4 (0.5%)	2.29	10.4 (0.1%)	0.37	262.9 (2.5%)	4.68
Poland	10394.4	1670.12	281.8 (2.5%)	11.8	212 (1.9%)	7.35	792.0 (7.1%)	36.14
Portugal	2055.7	146.47	55.4 (2.4%)	1.97	12 (0.5%)	0.37	229.8 (10.1%)	6.70
Romania	7916.9	1508.16	140.4 (1.7%)	3.84	36.3 (0.4%)	0.76	351.8 (4.3%)	15.87
Russia	176838.2	10422.40	863.3 (0.4%)	163.6	644.3 (0.3%)	119.17	3342.1 (1.9%)	302.4
Serbia	3515.7	469.46	52.4 (1.4%)	1.78	30.9 (0.8%)	0.75	179.6 (4.9%)	6.99
Slovakia	2256.3	437.99	38.1 (1.6%)	1.43	21.6 (0.9%)	0.81	99.1 (4.2%)	4.84
Slovenia	1292.2	282.25	11.1 (0.8%)	0.47	1.9 (0.1%)	0.11	33.2 (2.5%)	2.07
Spain	13246.1	1112.74	139 (1.1%)	5.42	113.4 (0.9%)	3.69	945.1 (6.7%)	27.09
Sweden	27315.1	2310.19	113.5 (0.4%)	5.58	26 (0.1%)	0.77	526.2 (1.9%)	9.79
Switzerland	1302.1	279.21	43.2 (3.6%)	2.1	3.7 (0.3%)	0.17	88.7 (6.4%)	4.31
United Kingdom	2687.5	238.00	317.5 (9.0%)	12.14	137.2 (3.9%)	4.39	762.3 (22.1%)	26.29
Ukraine	10657.6	1481.00	513.8 (4.2%)	17.01	461.4 (3.8%)	16.9	1377.6 (11.4%)	53.20

610 **Extended Data Table.1: PlanetScope based prediction statistics at country level.** Percentage is the contribution to the total
611 tree cover.



Supplementary Fig. 1: Overview of the model training process. We trained two models for tree segmentation and tree height regression.



Supplementary Fig. 2: Mismatch in training data. a, PlanetScope Imagery in 2019. **b,** Airborne LiDAR CHM data in 2017. **c,** Overlap between the two data sources; the high contrast area represent a mismatch due to forest logging and management.